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DECOUPLED REPROJECTION CONSISTENCY FOR DIAGNOSING 3D GAUSSIAN SPLATTING FAILURES

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One-line tagline

Training-free, coverage-aware triage for separating geometry/visibility failures from view-dependent instability in 3D Gaussian Splatting.

PROBLEM

Reprojection consistency—warping one view into another and measuring photometric agreement—is a standard sanity check for radiance-field reconstructions [WBF*17], but a single reprojection error map is ambiguous. Elevated error can arise from geometry/visibility failures, such as depth errors and occlusions, or from view-dependent appearance, such as highlights and reflections. In 3DGS, these effects are entangled because view dependence is encoded by spherical harmonics [KKLD23], so single-map diagnostics are useful for ranking but weak for debugging. [STGP26,GRS*24]

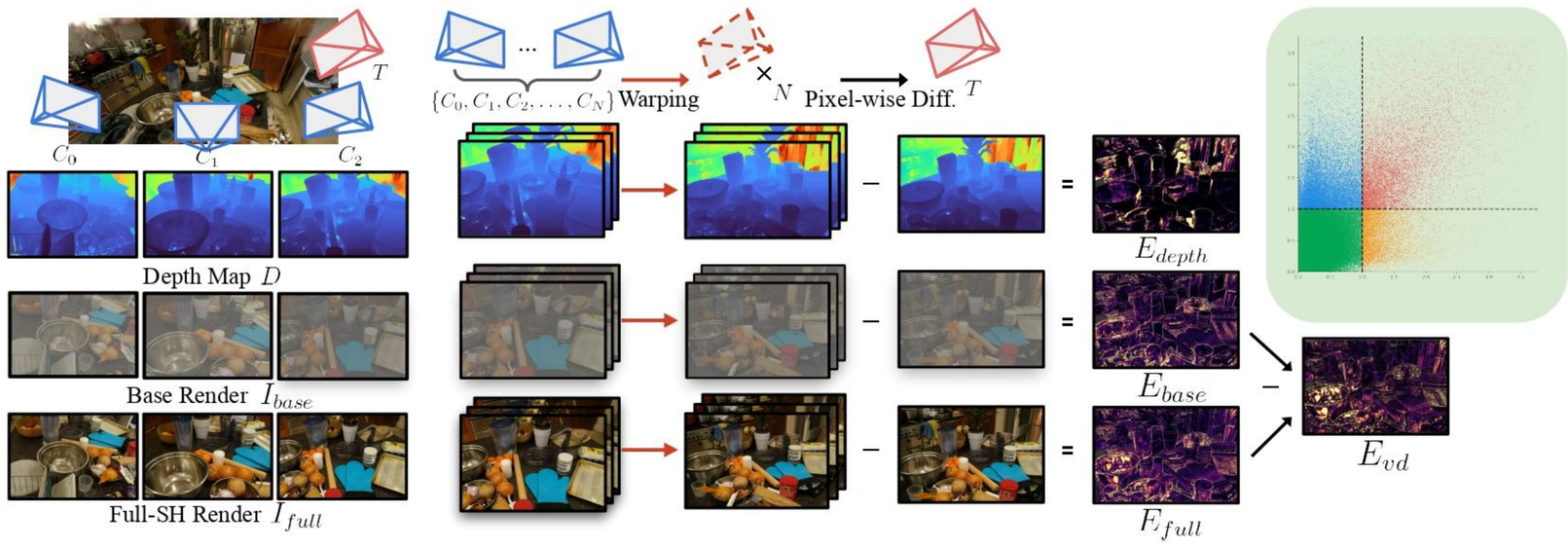
RELATED WORK

Existing training-free consistency diagnostics typically provide a single scalar map [STGP26,GRS*24]. DRC keeps the workflow training-free while adding a cause-separating axis tailored to 3DGS debugging.

KEY IDEA / OVERVIEW

DRC uses a simple internal ablation of the same trained 3DGS model. We compare reprojection inconsistency from a base render with SH truncated to L=0 and a full render with the trained SH degree. This yields three interpretable signals: E_{depth}, E_{base}, and $E_{vd} = max(0, E_{full} - E_{base})$.

METHOD OVERVIEW



Reprojection. For a target pixel x in view T, the rendered depth $D_T(x)$is used to lift x to 3D. The resulting 3D point is then projected into a neighboring view C_j, and the neighbor image is bilinearly sampled at the projected coordinate $\tilde{x}$. The sampled value is used as the warped neighbor value at pixel x in T and compared with the target pixel.

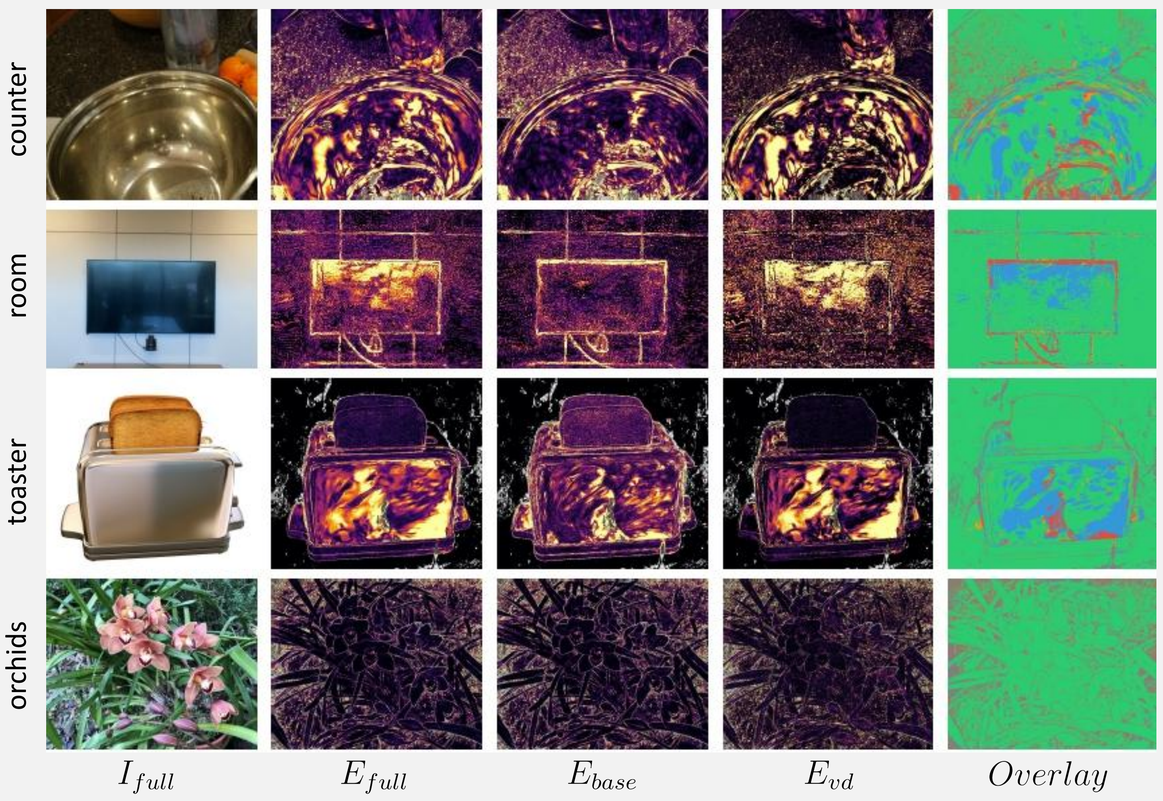
Decoupling. The same cross-view reprojection is performed twice on the same trained 3DGS model: once with SH truncated to $L = 0$, and once with the full trained SH degree. This produces E_{base}, E_{full}, and the view-dependent residual $E_{vd} = max(0, E_{full} - E_{base})$, while E_{depth} measures depth reprojection mismatch.

Validity and output. E_{depth} is evaluated on M_{geo}, whereas E_{base} and E_{full} are evaluated on M_{photo}. Photometric coverage C_{photo} is also reported. After shared per-view normalization, each valid pixel is labeled as Consistent, Geo/Vis-like, Appearance-driven, or Debug/Unexpected.

Scope note. $L = 0$ is used only as an internal SH truncation for diagnostics; it does not represent a physical diffuse or albedo decomposition.

RESULTS

DRC separates glossy/view-dependent regions from geometry/visibility failures across both synthetic and real scenes. The single map E_{full} conflates these effects, whereas E_{vd} isolates reflective behavior and the quadrant labels provide interpretable per-pixel triage.



Scene (N)	$\bar{C}_{\text{photo}}$	Cons.	Geo	App	Dbg
M360/counter (216)	0.947	85.8	5.8	6.1	2.2
LLFF/room (37)	0.967	85.1	5.0	8.1	1.8
Syn/toaster (100)	0.854	85.0	5.0	8.2	1.8
LLFF/orchids (23)	0.852	87.7	7.7	3.1	1.4

ROIs in COUNTER and ROOM are used only for visualization; all statistics are computed on full target views.
Mean photometric coverage remains high across scenes (0.85–0.97), and the appearance-driven fraction tracks scene glossiness.

Quadrant overlay
🟩 **Consistent** — both low
🟦 **Appearance-driven** — low E_{base}, high E_{full}
🟥 **Geo/Vis-like** — both high
🟧 **Debug/Unexpected** — high E_{base}, low E_{full}

Scene callouts
counter: Marble raises both E_{full} and E_{base}, while the stainless bowl is emphasized by E_{vd}.
room / toaster: Glossy surfaces are appearance-driven, while structural edges fall into the Geo/Vis-like regime.
orchids: Appearance-driven pixels are sparse in a mostly matte scene.

LIMITATIONS / TAKEAWAY

DRC is a diagnostic heuristic, not a ground-truth classifier. Very sharp/high-frequency reflections may be partially absorbed into low-order components (“baked-in”), reducing separability under SH truncation. Low coverage yields sparse maps, so C_{photo} should always be reported. Despite these caveats, DRC turns an ambiguous single error map into a practical, training-free, coverage-aware debugging signal for 3DGS. Future directions include reflection-aware decoupling beyond SH truncation and robust coverage-aware diagnostics for harder real-world capture conditions.

Author / Contact

Website
Project Page

Open to research internships & collaboration

REFERENCES

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